

ME23312 ENGINEERING THERMODYNAMICS

Unit I – Basics, Zeroth and First Law · Presentation 8 of 10

Enthalpy, Specific Heats and the First Law Applied to Non-flow Processes

Enthalpy · constant-pressure heating · c_v and c_p · the ideal-gas relations · the first-law table for the standard processes · multi-process cycles

In this presentation

1. Enthalpy
2. Heat added at constant pressure
3. Specific heats c_v and c_p
4. Ideal gas: $du = c_v dT$ and $dh = c_p dT$

1. The relation $c_p - c_v = R$
2. First-law table for the standard processes
3. Heat transfer in a polytropic process
4. Multi-process cycles – the first-law table method

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Enthalpy

Enthalpy – a new property

$$H = U + pV \quad (\text{kJ}) \qquad h = u + pv \quad (\text{kJ/kg})$$

- The combination $U + pV$ occurs naturally in constant-pressure heating and in **all flow processes**
- Enthalpy is a **property**, because U , p and V are properties
- For an ideal gas $h = u + RT$, so h also depends on temperature alone

Heat added at constant pressure

- Constant-pressure quasi-static process in a closed system: $W = p (V_2 - V_1)$
- First law: $Q = \Delta U + p (V_2 - V_1) = (U_2 + pV_2) - (U_1 + pV_1)$
- Therefore **$Q = \Delta H$** : the heat added at constant pressure equals the increase in enthalpy

KEY POINT At constant volume $Q = \Delta U$; at constant pressure $Q = \Delta H$.

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Specific heats

Specific heats c_v and c_p

$$c_v = (\partial u / \partial T)_v \quad c_p = (\partial h / \partial T)_p \quad (\text{kJ/kg K})$$

- c_v – heat to raise unit mass by one degree at constant volume; no displacement work, so $\delta Q = dU$
- c_p – heat to raise unit mass by one degree at constant pressure; $\delta Q = dH$

Ideal gas: u and h depend on T only

- The partial derivatives become ordinary derivatives
- $du = c_v dT$ and $dh = c_p dT$ for *any* process – not only at constant volume or constant pressure
- With constant specific heats: $\Delta U = mc_v (T_2 - T_1)$ and $\Delta H = mc_p (T_2 - T_1)$

KEY POINT For an ideal gas $\Delta U = mc_v \Delta T$ even in an isobaric, polytropic or adiabatic process.

Relation between c_p , c_v and R

$$h = u + RT \Rightarrow dh/dT = du/dT + R \Rightarrow c_p - c_v = R$$

$$\gamma = c_p/c_v, \quad c_v = R/(\gamma - 1), \quad c_p = \gamma R/(\gamma - 1)$$

- $c_p > c_v$: at constant pressure part of the heat goes into expansion work $p dV$
- Air: $R = 0.287$, $c_p = 1.005$, $c_v = 0.718$ kJ/kg K, $\gamma = 1.4$;
monatomic gases $\gamma = 1.67$

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The first law applied to non-flow processes

$Q = \Delta U + W$ for a mass m of ideal gas with constant specific heats

First-law table for the standard processes

Process	Law	Work W1–2	ΔU	Heat Q1–2
Constant volume	$V = C$	0	$mc_v (T_2 - T_1)$	$\Delta U = mc_v (T_2 - T_1)$
Constant pressure	$p = C$	$p (V_2 - V_1) = mR (T_2 - T_1)$	$mc_v (T_2 - T_1)$	$\Delta H = mc_p (T_2 - T_1)$
Isothermal	$pV = C$	$p_1 V_1 \ln (V_2/V_1)$	0	$Q = W$
Polytropic	$pV^n = C$	$(p_1 V_1 - p_2 V_2)/(n - 1)$	$mc_v (T_2 - T_1)$	$\Delta U + W = W (\gamma - n)/(\gamma - 1)$
Reversible adiabatic	$pV^\gamma = C$	$(p_1 V_1 - p_2 V_2)/(\gamma - 1) = -\Delta U$	$mc_v (T_2 - T_1)$	0

Heat transfer in a polytropic process

$$Q = W (\gamma - n)/(\gamma - 1)$$

- For $1 < n < \gamma$ (compression with cooling) the heat is **negative** during compression – part of the work is rejected as heat
- The process then lies **between** the isothermal and the adiabatic curves on the p–V diagram
- $n = 1$ gives $Q = W$ (isothermal); $n = \gamma$ gives $Q = 0$ (adiabatic)

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Multi-process cycles

The first-law table method

Solving a first-law table

1. For each process with two known quantities, find the third: $Q = \Delta E + W$
2. Use $\sum \Delta E = 0$ over the cycle to find the missing ΔE
3. Use $\sum Q = \sum W$ (or the given cyclic total) to find the missing Q or W
4. Check: $\sum W \text{ column} = \sum Q \text{ column}$

KEY POINT Worked Example 1.1 in the notes applies this method to a four-process cycle.

Cycle relations to remember

For any closed-system cycle: $\Sigma Q = \Sigma W$ and $\Sigma \Delta E = 0$

- Net work of a cycle = net heat transfer; positive net work $\Rightarrow$ heat engine, negative $\Rightarrow$ work-absorbing machine
- For a cycle of standard processes, find each W from the p – V formulas, then $Q_{\text{net}} = W_{\text{net}}$

Summary

- Enthalpy $H = U + pV$, $h = u + pv$ – a property; $Q = \Delta H$ at constant pressure
- $c_v = (\partial u / \partial T)_v$, $c_p = (\partial h / \partial T)_p$
- Ideal gas: $du = c_v dT$, $dh = c_p dT$ for any process
- $c_p - c_v = R$; $\gamma = c_p / c_v$; $c_v = R / (\gamma - 1)$
- Isochoric $Q = \Delta U$; isobaric $Q = \Delta H$; isothermal $Q = W$; adiabatic $Q = 0$
- Polytropic $Q = W (\gamma - n) / (\gamma - 1)$
- Cycle: $\Sigma Q = \Sigma W$, $\Sigma \Delta E = 0$ – the first-law table method

Next: The steady flow energy equation

- Control volume and flow work
- Assumptions of steady flow and the derivation of the SFEE
- Meaning and magnitude of each term
- Numerical problems: Worked Examples 1.1 – 1.5 in the lecture notes